TUM Institute for Cognitive Systems (ICS)
Multi-Sensory Based Robot Dynamic Manipulation

Tutorial: How to use the Schunk Net box and Force-Torque Sensor

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This tutorial explains how to start and read the Schunk force-torque sensor mounted on the UR 10 robot in the lab. The sensor uses a Ethernet connection with Power over Ethernet (PoE+) feature. Do not touch the connection of the robot or the PC cable.

IMPORTANT: If you notice anything strange with the sensor or its cable, do not operate the robot and contact one of the Course supervisors.

1 Verify connections

Before start working with the robot, connect the green Ethernet cable of the Schunk Net-Box in the first port of the PoE+ switch as shown in fig 1.1. After that, you will notice some blinking LEDs inside the Box. The sensor will take around 1 minute to start.

Once the sensor is ready, make sure that your PC is connected to the robot network. The IP of the sensor is **192.168.1.1**. Verify that you can ping its address from a terminal.

```
ping 192.168.1.1
```

If your see feedback on the terminal the sensor is ready.

2 Calibrate sensor

Once the connection with the sensor is established, you can open the web commander in your favorite web browser. Enter the IP of the sensor in the URL bar of your browser. You will land in the welcome page of the Schunk Net-box web commander as shown in figure 2.1. Then, click on **Snapshot**.

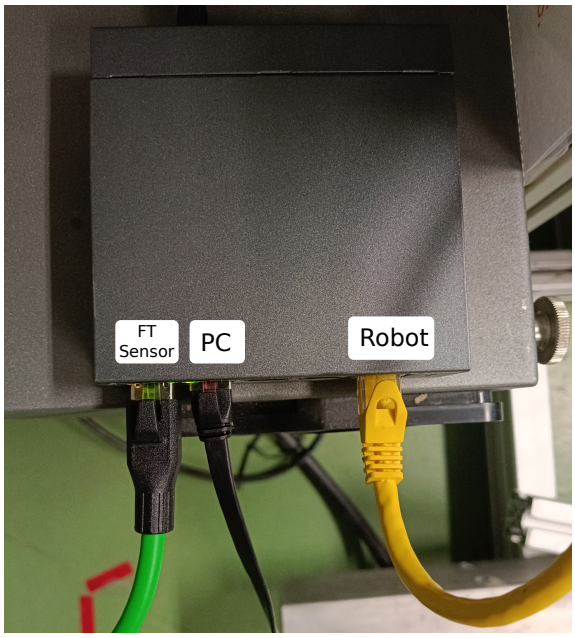


Figure 1.1: Connections in the robot switch.

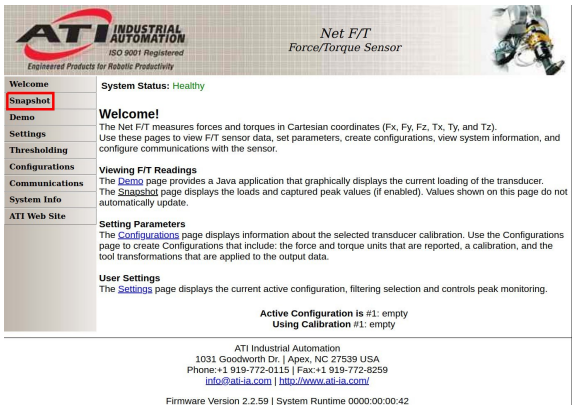


Figure 2.1: Welcome page of the Schunk Net-box web commander.

The sensor by default provides raw measurements as shown in figure 2.2. You can calibrate zeroes by clicking on **Bias**, and then on **Reset Peaks**.

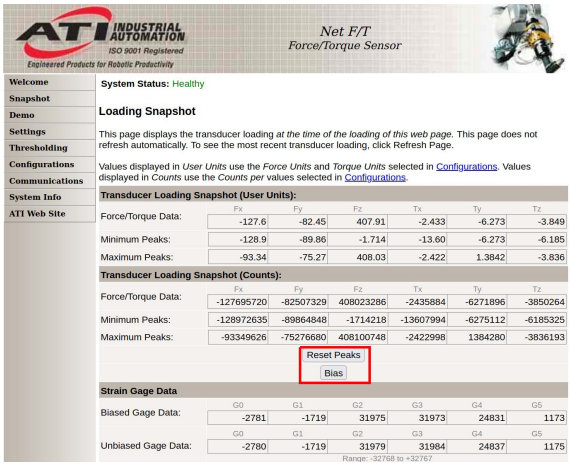


Figure 2.2: Default raw readings from the sensor may be very high. You must calibrate them.

Now the readings of the sensor should be close to zero.

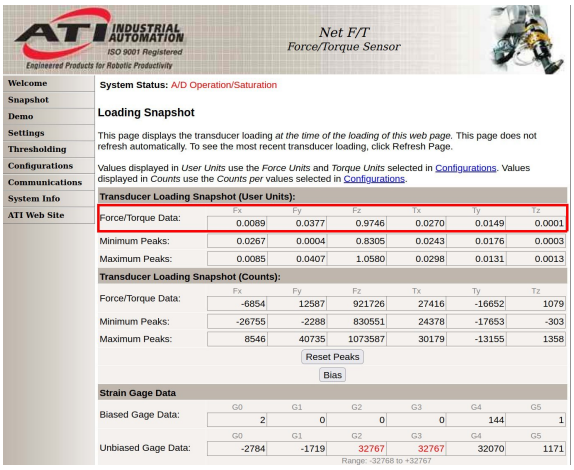


Figure 2.3: After calibration, the readings of the sensor should be close to zero.

3 ROS node

Once the sensor is ready and calibrated, you can add the `tum_ics_schunk_netbox` package to the `src` folder of your catkin workspace and compile it again.

Then you can launch the `sensor_publisher` node with the following command.

```
roslaunch tum_ics_schunk_netbox sensor_publisher.launch
```

The node publishes the sensor data in the topic `/schunk_netbox/raw`. It is published in a stamped wrench on the link `ft_sensor_link`. You can test it by echoing the topic.

```
rostopic echo /schunk_netbox/raw
```

You can visualize the applied wrench by loading the rviz pluging for stamped wrenches.